Department of Computer Science (FCA) - Ganpat University						
B.Sc. IT (CA & IT) Hons. - Semester III : Assignment						
Subject :U13C4AI: INTRODUCTION TO ARTIFICIAL INTELLIGENCE						
BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create;						
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Unit Number	**Sr. No.**	**Questions**	**Name of CO**	**Matching Program Outcome (PO)**	**Bloom's Taxonomy (BTL)**	**Mini Project / Case Studies**
Basics of Artificial Intelligence:	1	Explain Artificial Intelligence and discuss its advantages, disadvantages, and applications.	CO 1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL2	
	2	Compare the different types of Artificial Intelligence with suitable examples.	CO 1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	3	Explain AI Agents and discuss different types of AI Agents.	CO 1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL2	
	4	Explain major AI techniques used in Artificial Intelligence systems.	CO 1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL2	
	5	Analyze the role of AI in modern industries with real-life examples.	CO 1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	6	Differentiate between Human Intelligence and Artificial Intelligence.	CO 1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	7	Explain the importance of AI and discuss its future scope.	CO 1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL2	
Problems, State Space Search & Heuristic Search Techniques:	1	Analyze how state space search can be applied to solve a real-world problem and evaluate its effectiveness.	CO 2	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	2	Compare production systems and state space search approaches for AI problem-solving with suitable examples.	CO 2	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	3	Assess the major issues involved in designing search programs and their impact on search efficiency.	CO 2	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	4	Evaluate the strengths and limitations of the Generate-and-Test and Hill Climbing techniques in solving complex problems.	CO 2	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	5	Compare Best-First Search and Problem Reduction techniques in terms of solution quality and computational efficiency.	CO 2	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	6	Analyze how Constraint Satisfaction techniques can be used to solve scheduling and resource allocation problems.	CO 2	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	7	Evaluate the role of Means-Ends Analysis in goal-directed problem-solving and compare it with other heuristic search techniques.	CO 2	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
Introduction to Knowledge representation:	1	Analyze the role of knowledge representation and mappings in enabling intelligent decision-making in AI systems.	CO 3	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	2	Compare different approaches to knowledge representation and evaluate their suitability for real-world AI applications.	CO 3	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	3	Assess the major issues in knowledge representation and propose strategies to overcome them in knowledge-based systems.	CO 3	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL6	
	4	Analyze how simple facts, instance relationships, and IS-A relationships can be represented using logic-based knowledge representation techniques.	CO 3	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	5	Evaluate the significance of computable functions, predicates, and resolution in automated reasoning systems.	CO 3	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	6	Compare natural deduction and resolution methods for logical inference and assess their effectiveness in problem-solving.	CO 3	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	7	Analyze the differences between procedural and declarative knowledge and evaluate their roles in logic programming and AI reasoning.	CO 3	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	8	Compare forward reasoning and backward reasoning techniques and justify their use in expert systems and intelligent agents.	CO 3	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
Symbolic reasoning under uncertainty, Statistical Reasoning, Weak slot and filler structures:	1	Analyze the role of symbolic and statistical reasoning in handling uncertainty within intelligent systems.	CO 4	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	2	Compare non-monotonic reasoning, default reasoning, and minimal reasoning in dynamic decision-making environments.	CO 4	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	3	Evaluate the effectiveness of dependency-directed and non-dependency-directed backtracking techniques in problem-solving.	CO 4	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	4	Assess the role of Truth Maintenance Systems (JTMS and LTMS) in maintaining consistency in knowledge-based systems.	CO 4	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	5	Compare Bayesian reasoning and Dempster-Shafer theory for reasoning under uncertainty and justify their applications.	CO 4	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	6	Analyze the structure and significance of Bayesian Networks in representing probabilistic relationships among variables.	CO 4	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	7	Evaluate the effectiveness of Semantic Nets, Partitioned Semantic Nets, and Frames for knowledge representation in AI systems.	CO 4	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
Strong slot and filler structures, Parallel and distributed AI, Natural language processing:	1	Analyze the role of Conceptual Dependency, Scripts, and CYC in representing and understanding complex knowledge in AI systems.	CO 5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	2	Compare strong slot-and-filler structures such as Frames, Scripts, and Conceptual Dependency in terms of knowledge representation efficiency.	CO 5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	3	Evaluate the contribution of psychological modeling to the development of intelligent and human-like AI systems.	CO 5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	4	Assess the advantages and challenges of parallel reasoning systems and distributed reasoning systems in solving large-scale AI problems.	CO 5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	5	Compare parallel AI and distributed AI approaches with respect to performance, scalability, and knowledge sharing.	CO 5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL4	
	6	Analyze the stages of Natural Language Processing and evaluate the role of syntactic, discourse, and pragmatic processing in language understanding.	CO 5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	
	7	Evaluate how knowledge representation structures and NLP techniques work together to improve human-computer interaction and intelligent decision-making.	CO 5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,	BTL5	